

US Tariff Impact Tracker: Imports into LA Mixed Over Next Two Weeks on a YOY Basis

US Tariff Impact Tracker: This past week, laden vessels from China to USA were down sequentially (-3% WoW) and down on a YoY basis (-14.5% YoY). Data suggests TEUs coming into Port of LA will turn positive next week (+2% WoW) following this past week's +2% sequential move, before seeing a similar increase of +2% WoW two weeks out (YoY is expected to be +8.5% and then -10%, one week and two weeks out). What we will need to monitor is how levels move through July and August, which could suggest how shippers are deciding to restock, how early peak could be starting (and whether there was an early start to peak shipping this spring), and with in some cases how lower effective tariff rates impact import decisions amidst an uncertain geopolitical backdrop.

Trade uncertainty remains prevalent in global trade.

Jordan Alliger
+1(212)357-4913 |
jordan.alliger@gs.com
Goldman Sachs & Co. LLC

Paul Stoddard
+1(801)744-3761 |
paul.x.stoddard@gs.com
Goldman Sachs & Co. LLC

Andrzej Tomczyk, CFA
+1(212)357-4445 |
andrzej.tomczyk@gs.com
Goldman Sachs & Co. LLC

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

Key observations from this past week's report:

- **China freight flows (week ending Thursday, July 23rd) showed a sequential decline in laden vessels from China to the US (-3%) with a negative YoY at -14.5% (vs -18% in the prior week).**
- **Port Optimizer sequentials indicate imports into the Port of Los Angeles are set to be up +2% TEUs (July 31) while the early read suggests an increase of +2% on a two-week out basis;** on a YoY basis, imports into LA are set to trend positive to +8.5% and then decrease -10% two-week out.
- **Rail intermodal volumes along the West Coast were up +8% YoY following last week's +4% performance.**
- **Ocean container rates were down -12% sequentially; rates on a YoY basis were up +167% YoY this past week.** We anticipate some choppiness could remain over the coming weeks as global capacity potentially shifts given ongoing geopolitical events.
- **Load availability for trucks on the West Coast increased last week sequentially (+2%) and was negative on a YoY basis (-5%); truck spot rates on the West Coast were up +56% YoY ex-fuel.**

What the Tracker Is:

What We Disseminate Weekly: High frequency data to help assess the ongoing impact of tariffs on global supply chains and the accompanying ramifications on the flows of freight (e.g., expected ships leaving from China to USA). While we think our data set is representative, we do plan to periodically add new data series as they become available.

It is important to recognize that weekly data can be volatile, with a fair bit of noise depending on timing—but we still think looking at the data collectively, and perhaps over a multi-week basis can be informative as to tariff-related trends.

2026 Trade and Transport Scenario Roadmap—Revised From First and Second Edition:

As we moved past the 2025 homestretch and into 2026, the ability to find a profit and earnings bottom and eventually see an earnings upgrade cycle comes down to one thing: volume growth, with an appropriate weighting towards higher margin business-to-business, commercial and/or manufacturing flows. Recent improved share price performance from the trucking sector could perhaps be tied to the notion that volumes (and in terms of truckload - supply) could indeed be at a more stable point with which improvement could come from at some point in 2026.

Ramifications from tariff-related uncertainty that caused pull-forward in demand—bracketed by shipper indecision as to how much product to manufacture and/or order for inventory—had been a culprit behind transports broad underperformance for much of 2025 and tied directly into our view that we may see subseasonal peak season shipping demand (for 4Q reporting).

That said, we remain positive on the cycle recovery story—despite the heretofore elusive bottom, and we note several factors that could finally stimulate a favorable and fulsome volume inflection into 2026 as we look to the medium to longer term.

- Fed rate cut cycles are typically beneficial to transport shares (see [Fed Rate Cuts and Transports](#)); GS economists predict one rate cut in 2026 (December) and another cut in March 2027 to add to the three cuts in 2025
- While tariff uncertainty still exists, we lapped Liberation Day on April 2, 2026—potentially yielding a more consistent playbook for shippers to work with and to plan.
- Various corporations are announcing increased investments in US manufacturing—including companies like Apple, Nvidia, IBM, Pfizer, Johnson & Johnson, etc.—which should auger-in more domestic freight flows.
- The bonus depreciation revival via the *One Big Beautiful Bill Act* could incentivize capital to reinvest in businesses.
- While tariff deployment is disruptive, any resulting re-shoring or nearshoring would bolster US manufacturing, which also should auger-in more domestic freight flows.
- Shifting logistics patterns and changing supply chain sourcing could drive secular global trade opportunity as companies potentially pursue a China Plus 1 or 2 strategy.

What to do with Transport stocks:

- We upgraded the truckers last year ([see our upgrade note](#)), as we believed the likelihood of recession had lessened, and as the consumer has stayed relatively resilient. While freight forwarders such as EXPD and CHRW could benefit from the volatility and surge in customs brokerage demand, we do note that YoY comparisons for ocean rates will be challenging. Should the Red Sea re-open, that could further add to effective capacity.
- **Other beneficiaries could be the parcels (UPS and FDX—both Buy-rated) with their fast cycle logistics and air freight as well as large global footprints that can help shippers shift supply chains.**

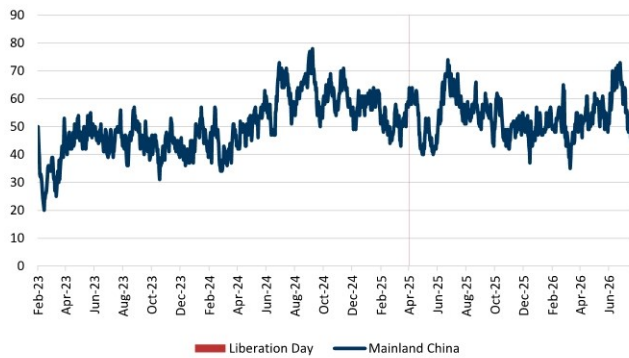
High Frequency Weekly Data

We track weekly (and some daily) data that helps provide as real time as possible insights into trade volumes, demand and pricing. We note that much of this data is choppy in the near term and more than often shows trends developing over time as we receive more data, but also speaks to the volatility of trade flows. Therefore, we caution against drawing conclusions week to week but note that it can be helpful to see changes in the data week to week given the dynamic trade environment. Below, we track the high frequency data points that we believe are highly relevant to trade and US freight transportation.

- **Laden container vessels from China to the US were -14.5% YoY over the last week (7/17-7/23) versus -18% YoY in the prior week (7/10-7/16). The sequential vessel decrease of -3% in the most recent week follows last decrease of -15%.**

Exhibit 1: Container ships to the US remain volatile; levels are nearing post-Liberation day increases

Daily Laden Container Ship Vessels from China to US

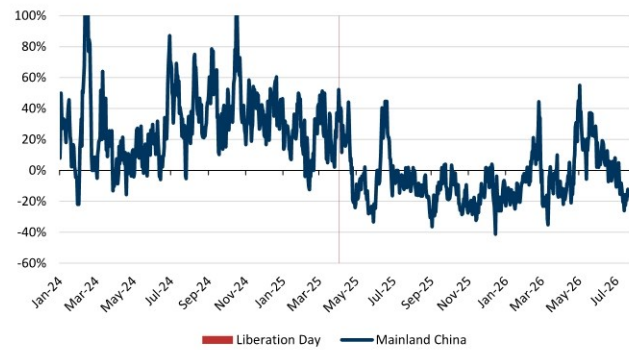


Data through 7/23/26 Captures the number of laden (i.e., cargo-carrying) container ships departing China en route to the US over a rolling 15-day window. Includes only vessels confirmed to be transporting cargo. Laden vessels are those reporting a draft of more than 75% of maximum draft at the start of their journey to the US. Note that data is subject to revisions.

Source: Bloomberg

Exhibit 2: Laden container ships from China to US have accelerated on a YoY basis in the most recent week

Daily Laden Container Ship Vessels from China to US, YoY



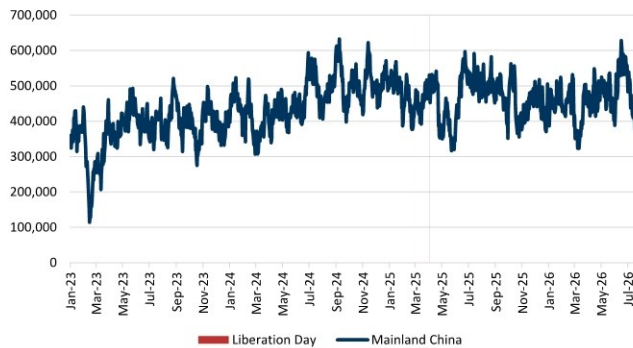
Data through 7/23/26 captures the number of laden (i.e., cargo-carrying) container ships departing China en route to the US over a rolling 15-day window. Includes only vessels confirmed to be transporting cargo. Laden vessels are those reporting a draft of more than 75% of maximum draft at the start of their journey to the US. Note that data is subject to revisions.

Source: Bloomberg

- **TEUs from China to the US were -23% YoY over the past week (7/17-7/23) versus -18.5% in the prior week (7/10-7/16). Sequentially, we saw a -7% change in TEUs from Mainland China.**

Exhibit 3: TEUs have remained volatile

Daily TEU from China to the US

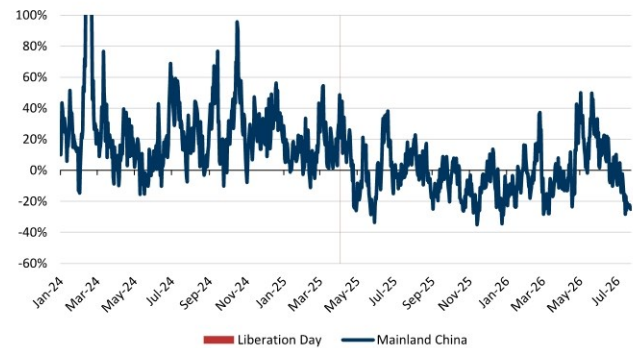


Data as of 7/23/2026. Represents the aggregated container volume, measured in twenty-foot equivalent units (TEU), of vessels departing China for the United States over a 15-day rolling period. Accounts for the shipping capacity being utilized, irrespective of the number of vessels. Dry cargo vessels leaving China are monitored as they exit Chinese waters. The dataset is filtered for vessels that have left over the 15 previous days and have reported the US as their country of destination. Note that data is subject to revisions.

Source: Bloomberg

Exhibit 4: TEU growth decelerated in the most recent week

Daily TEU from China to the US, YoY



Data as of 7/23/2026. Represents the aggregated container volume, measured in twenty-foot equivalent units (TEU), of vessels departing China for the United States over a 15-day rolling period. Accounts for the shipping capacity being utilized, irrespective of the number of vessels. Dry cargo vessels leaving China are monitored as they exit Chinese waters. The dataset is filtered for vessels that have left over the 15 previous days and have reported the US as their country of destination. Note that data is subject to revisions.

Source: Bloomberg

Exhibit 5: Week 29 China to US vessels and TEUs were down WoW

Weekly Average TEUs and Vessels from China to US

Bloomberg China to US 6 Week Dashboard Weekly Average TEUs & Vessels							
Week	23	24	25	26	27	28	29
2026 Vessels	63	67	70	65	60	51	49
WoW		6.1%	3.8%	-7.0%	-7.9%	-14.9%	-3.4%
2025 Vessels	58	64	70	65	64	62	57
2025 WoW		11.7%	8.7%	-7.4%	-1.8%	-2.2%	-7.8%
2026 YoY	9.7%	4.2%	-0.4%	0.0%	-6.3%	-18.4%	-14.5%
2026 TEUs	517,628	575,345	555,551	540,110	492,602	427,500	398,691
WoW		11.2%	-3.4%	-2.8%	-8.8%	-13.2%	-6.7%
2025 TEUs	459,259	532,618	559,219	525,660	510,667	524,435	516,271
2025 WoW		16.0%	5.0%	-6.0%	-2.9%	2.7%	-1.6%
2026 YoY	12.7%	8.0%	-0.7%	2.7%	-3.5%	-18.5%	-22.8%

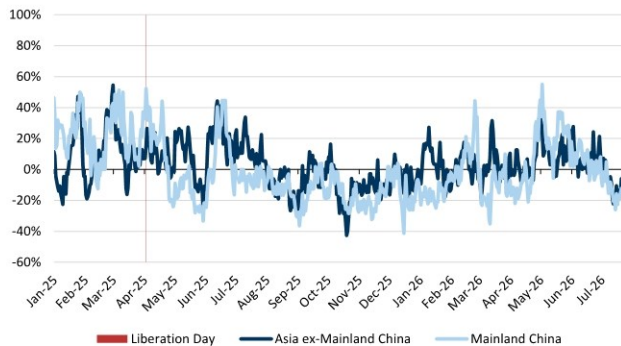
Week 29 7/17/2026-7/23/2026; Note that data is subject to revisions.

Source: Bloomberg

Mainland China's laden vessel and TEU growth vs. Asia Ex-Mainland China to US: On a laden vessel basis, Mainland China was down -15.5% YoY on average while Asia-Ex Mainland China was down -10.5% YoY. On a TEU basis, Mainland China was down -23% YoY on average while Asia-Ex Mainland was down -9% on average. We use the summation of Mainland China, Vietnam, South Korea, Taiwan and Japan as a proxy for Asia exports to the US.

Exhibit 6: Mainland China and Asia-Ex Mainland: vessels were down YoY for Mainland China and for Asia Ex Mainland China

Daily Laden Container Ship Vessels from Asia ex-Mainland China and Mainland China to US, YoY

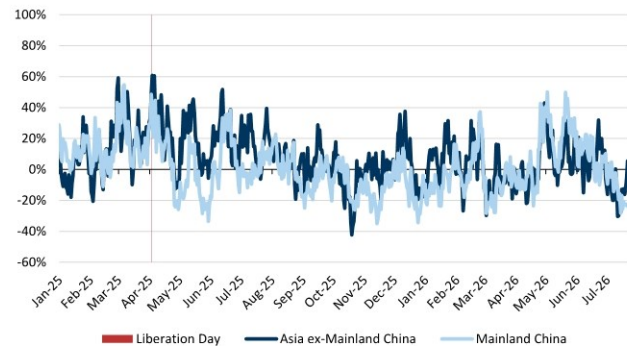


Asia ex-Mainland China includes Vietnam, South Korea, Taiwan, and Japan Data as of 7/23/2026

Source: Bloomberg

Exhibit 7: TEU growth from Mainland China decelerated Yoy and Asia-Ex Mainland China accelerated YoY this past week

Daily TEU from Asia ex-Mainland China and Mainland China to US, YoY

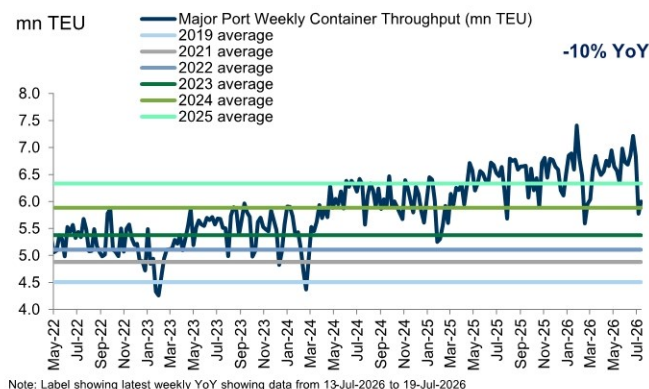


Asia ex-Mainland China includes Vietnam, South Korea, Taiwan and Japan Data as of 7/23/2026

Source: Bloomberg

- Chinese major port weekly throughput was down +4% w/w during the week ended 7/19 (latest available) after being down -15% w/w the prior week. **Throughput was -10% YoY vs -11% in the prior week.**

Exhibit 8: Chinese port activity was down +4% sequentially in the most recent week and down -10% YoY
Chinese major port weekly container throughput

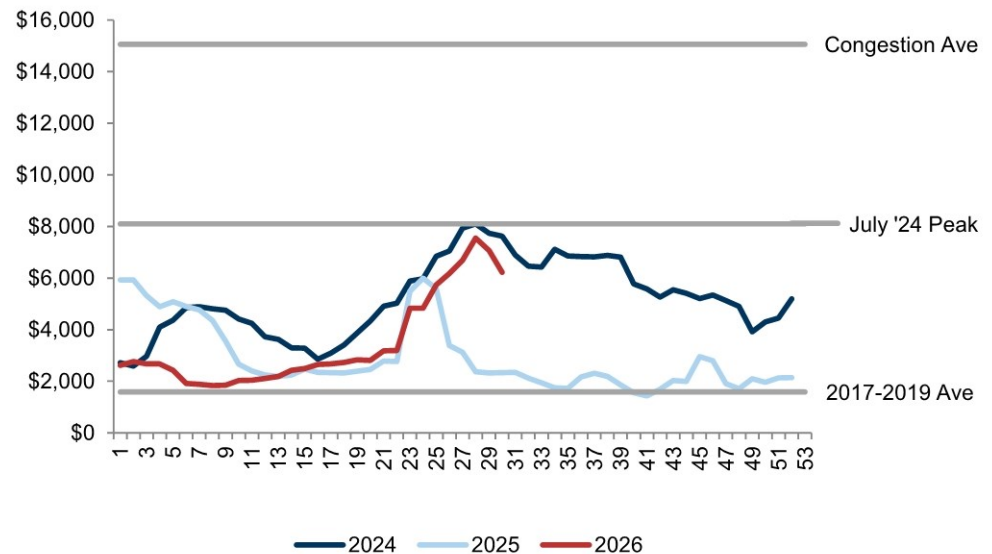


Weekly through 7/19/2026. Data compiled by our Asia Analysts, Herbert Lu, Simon Cheung and Dorothy Wong

Source: Ministry of Transport

- Ocean container rates from China/East Asia to the US West Coast were down -12% from last week and up +167% YoY in the most recent week. We anticipate some volatility over the coming weeks as global capacity shifts given ongoing geopolitical events, given potential surcharges, and considering peak season beginning earlier this year.

Exhibit 9: Ocean rates decreased -12% WoW after decreasing -6% in the previous week
China/East Asia to the North American West Coast, \$/FEU



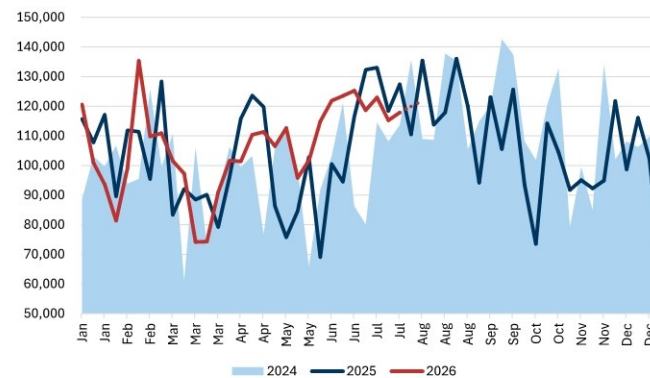
Weekly rate through 7/24/2026

Source: Freightos

- **Planned TEUs into the Port of LA in the most recent week were up +2% sequentially after +4%/-6% WoW moves in the prior two weeks. Looking ahead, this week's positive sequential move appears to remained unchanged next week and two weeks out. YoY trends are expected to increase +8.5% YoY for TEUs next week and then decrease -10% two weeks out.**

Exhibit 10: Planned TEUs are expected to stay unchanged next week and two weeks out

Planned TEUs into the Port of LA



Weekly data updated daily as of 7/24/2026 Port Optimizer data includes two week forward data for week ending 8/7

Source: Port Optimizer

Exhibit 11: Tracking volumes over the last four weeks and upcoming 2 weeks

Port Optimizer 6 Week Dashboard

Port Optimizer 6 Week Dashboard							
Date	6/26/2026	7/3/2026	7/10/2026	7/17/2026	7/24/2026	7/31/2026	8/7/2026
Week	25	26	27	28	29	30	31
TEUs	125,283	118,669	122,977	115,261	117,882	119,960	121,837
WoW	N/A	-5.3%	3.6%	-6.3%	2.3%	1.8%	1.6%
2025 WoW	23.3%	13.6%	0.6%	-11.0%	7.6%	-13.3%	22.6%
2026 YoY	7.5%	-10.3%	-7.6%	-2.7%	-7.5%	8.5%	-10.1%
Vessels	23	23	22	23	21	24	23
WoW	N/A	0.0%	-4.3%	4.5%	-8.7%	14.3%	-4.2%
2025 WoW	17.6%	20.0%	8.3%	-7.7%	-12.5%	4.8%	18.2%
2026 YoY	15.0%	-4.2%	-15.4%	-4.2%	0.0%	9.1%	-11.5%

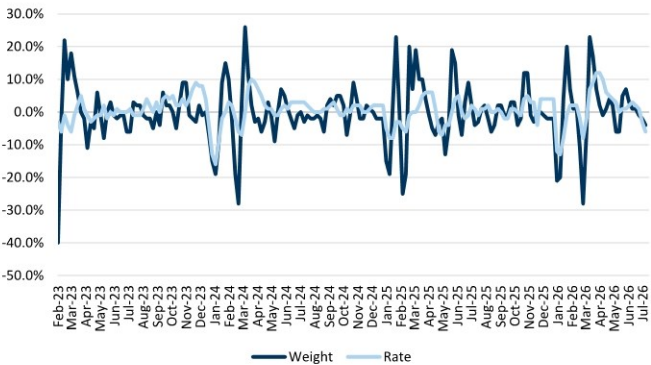
Weekly data update as of 7/24/2026 Port Optimizer data includes two week forward data for week ending 8/7

Source: Port Optimizer

- WorldACD Weekly Cargo Trends Asia Pacific to North America weights and rates (as of 7/16; latest), two week over two week was -4% and -6% last week, after -2% and -3% in the prior week, respectively. The sequential weight movements are figures to watch to ascertain the need for faster cycle logistics. **We continue to monitor rate impact on global and regional rates from reduced air capacity in Gulf states in addition to increase in jet fuel pricing, noting the limited visibility on weekly rates, as well as recognizing the dynamic situation as some airports have limited reopening of flights.**

Exhibit 12: Trends for air cargo out of Asia Pacific to North America: weight and rate increased in the most recent week (-4% and -6% respectively)

WorldACD Last Two Weeks Compared with the Preceding Two Weeks, Asia Pacific to North America Weight and Rate



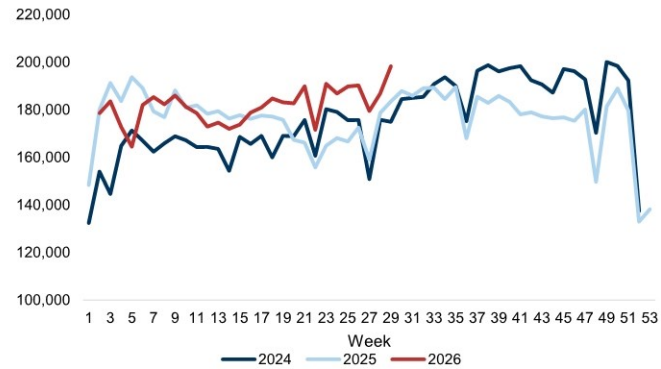
Weekly two week over two week change as of 7/16/2026

Source: WorldACD

- Intermodal traffic growth on the West Coast (UNP/BNSF) was +8% YoY in the most recent week, versus +4% last week.

Exhibit 13: West Coast intermodal carloads were +8% YoY versus +4% YoY last week

Intermodal Carloads on the West Coast



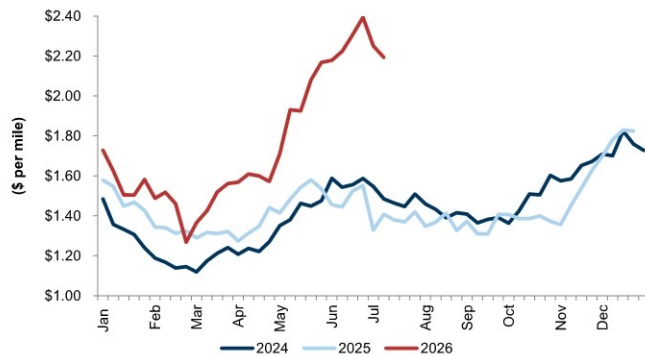
Weekly intermodal carload as of 7/18/2026

Source: AAR

- Truckload spot rates ex-fuel on the West Coast from truckstop.com increased -3% w/w and +56% YoY in the most recent week.
- Truck load availability index on the West Coast was up +2% w/w and up -5% YoY.

Exhibit 14: Rates were +56% YoY last week on the West Coast

Truckload Spot Rates ex-fuel West Coast

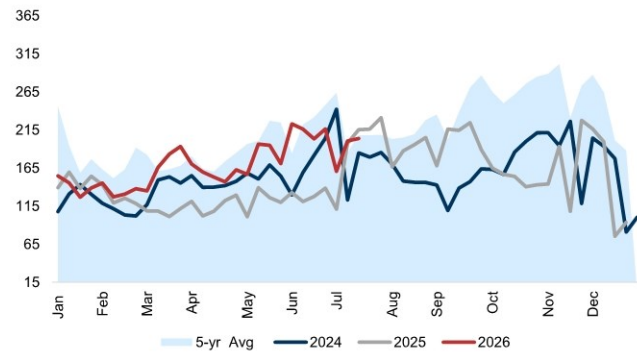


Weekly data as of 7/20/2026

Source: Truckstop.com

Exhibit 15: Truckload load availability was -5% YoY this past week on the West Coast

Truckload Load Availability Index on the West Coast



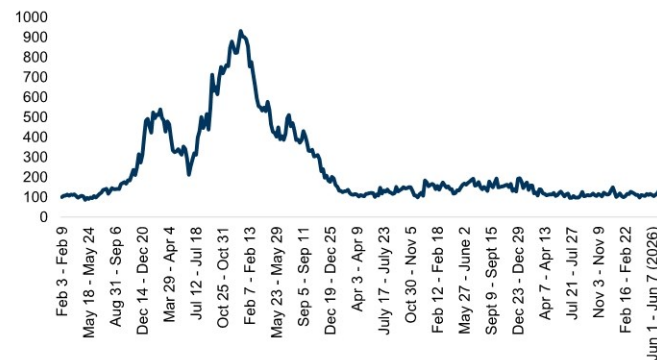
Weekly data as of 7/20/2026

Source: Truckstop.com

- Our Supply Chain Congestion Tracker stayed at '2' this past week as the bottleneck index was stayed flat w/w; overall fluidity levels are closer to the pre-Covid baseline.

Exhibit 16: Congestion remains fluid as per our supply chain congestion index, about in line with pre-Covid levels

GS Weekly Supply Chain Congestion Index, Feb 2020 - July 2026



Weekly data as of 7/12/2026

Source: Goldman Sachs Global Investment Research

Lagged Monthly Data

We include monthly data that we believe reflects the impact to freight flows from trade policy.

- Big Three (Ports of LA, Long Beach and Oakland) volumes were +30% YoY and up +2.5% sequentially from April to May—below historical seasonality of +12%.

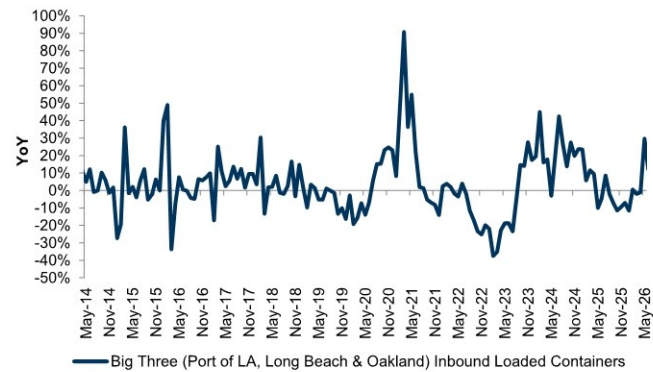
Exhibit 17: Volumes into the West Coast were up +2.5% in May from April, below historical seasonality

Big Three Volumes

May-2026	YoY%	Seq%	5-yr YoY% Avg	5-yr Seq% Avg
Port of Long Beach	40.0%	7.4%	-0.9%	12.1%
Port of LA	26.2%	-2.3%	4.2%	12.9%
Port of Oakland	5.7%	6.2%	2.6%	8.7%
Big Three Total	29.6%	2.5%	1.8%	12.0%

Source: Port of Los Angeles, Port of Long Beach, Port of Oakland

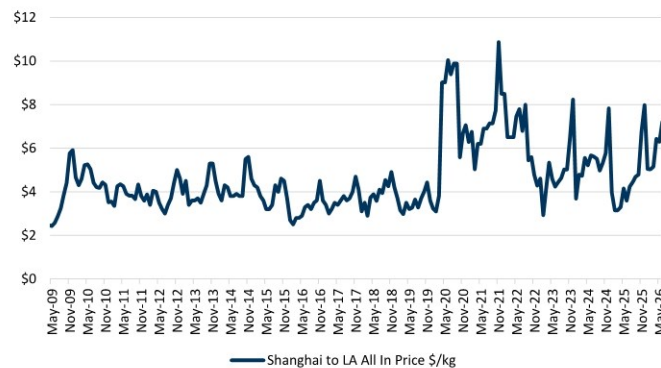
Exhibit 18: Imports into the West Coast were +30% YoY in May
Big Three Volumes YoY



Source: Port of Los Angeles, Port of Long Beach, Port of Oakland

- Drewry air rates from Shanghai to LA were down -2% MoM in May after being up +25% in April. Depending on the development of geopolitical events, we could see a subsequent increase of rates in June as air capacity gets tied up in addition to the rise in jet fuel prices—to be determined.

Exhibit 19: May rates were down -2% from April
Drewry Air Rates Shanghai to LA

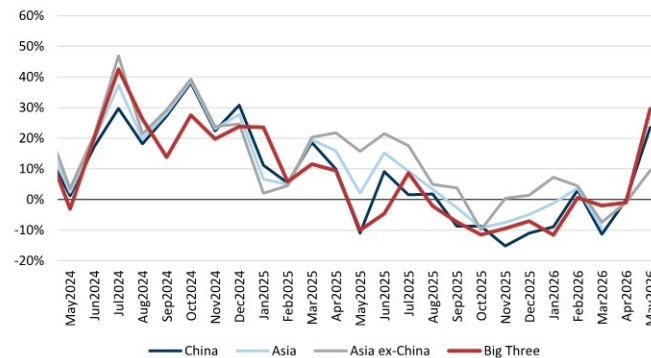


Source: Drewry

- **Big Three YoY v. Monthly YoY growth of Bloomberg reported TEUs from China, Asia and Asia ex-China to US shows a strong relationship.**

Exhibit 20: Big Three growth tends to follow TEU growth out of Asia to the US

Big Three YoY v. China/Asia/Asia ex-China to US TEU Monthly YoY



China and Asia TEU YoY data through 5/26

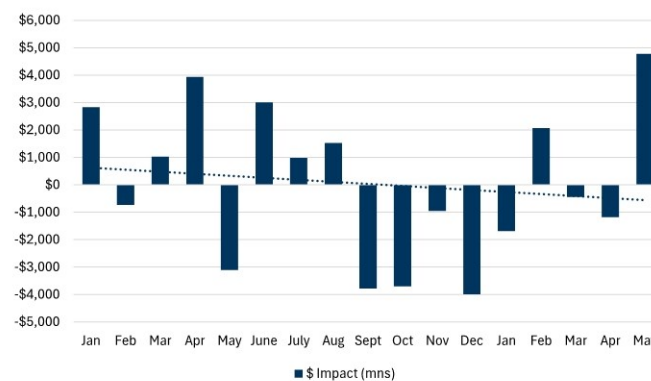
Source: Bloomberg, Port of Los Angeles, Port of Long Beach, Port of Oakland

- **We estimate that imports could have been up +\$4.8bn YoY in May following April's estimated YoY decrease of ~-\$1.2bn (Exhibit 21). This follows the prior month declines September through January.**

- We leverage a similar analysis from the Longshoreman strike last October (see our [longshoreman impact note](#)). According to the Bureau of Transportation Statistics, Ocean trade was about \$2.3tn in trade in 2022. The top 25 ports in the U.S. moved 44mn loaded TEUs in 2022, meaning that each loaded container (TEU) on average in 2022 had a value of ~\$52,000. Growing this by inflation of ~3% for the last three years implied value per container of ~\$57,000. Multiplying the YoY change in TEUs with the value/TEU, we estimate the monthly imported value change YoY.

Exhibit 21: We estimate that May levels were higher YoY

Implied Trade Value Change YoY Based on YoY TEU Change and Estimated Value/TEU



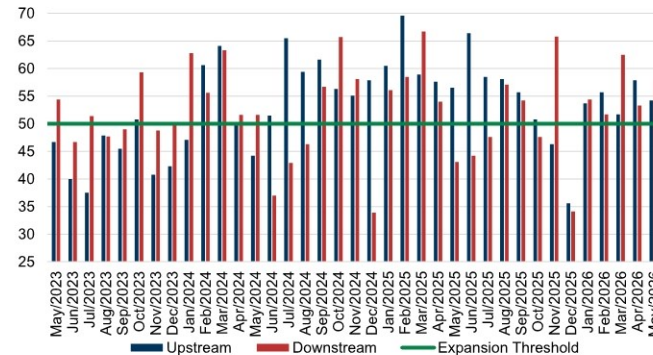
Source: Port Optimizer, US Bureau of Transportation Statistics (BTS), Goldman Sachs Global Investment Research

■ Logistics Managers Index Inventory Level Expansion

- May upstream (B2B) inventories expanded slower at 54.2 vs 57.9 in April.
- May downstream (Retail) inventories expanded faster at 57.8 after 53.3 in April.

Exhibit 22: LMI has shown upstream and downstream expansion

Logistics Manager Index—Inventory Level Expansion, Upstream (B2B) vs Downstream (Retail) Respondents



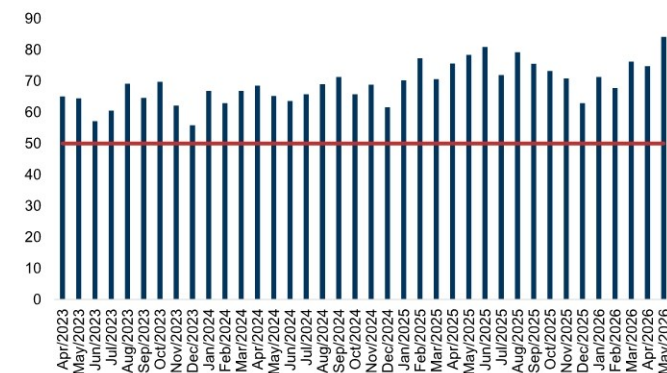
Source: Logistics Managers Index

■ Logistics Managers Index Inventory Cost Index

- Inventory cost index was 84.1 in May vs 74.7 in April, reflecting inventory costs expanding faster.

Exhibit 23: Inventory costs expanded faster in May vs April

LMI Inventory Cost Index

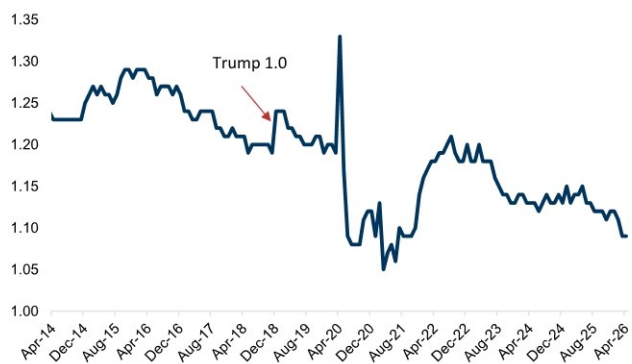


Source: Logistics Managers Index

- Inventory to sales for retailers/manufacturers/wholesalers were 1.09/1.50/1.19 for April which is lower versus March at 1.09/1.51/1.21.

Exhibit 24: Inventory to sales have not shown an increase like in Trump 1.0...

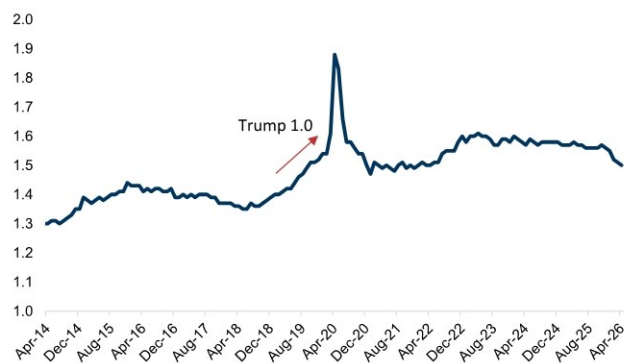
Inventory to Sales Ratio: Retailers ex Motor Vehicle and Parts Dealers



Source: Census Bureau

Exhibit 25: ...nor for manufacturers...

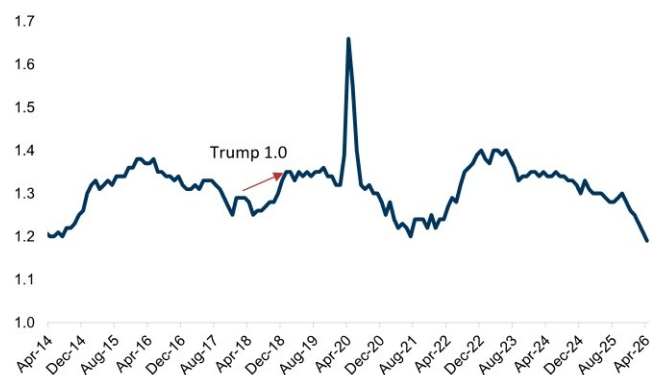
Inventory to Sales Manufacturers



Source: Census Bureau

Exhibit 26: ...while wholesalers were also unchanged

Inventory to Sales Wholesalers



Source: Census Bureau

Disclosure Appendix

Reg AC

We, Jordan Alliger, Paul Stoddard and Andrzej Tomczyk, CFA, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Jordan Alliger Goldman Sachs & Co. LLC, Paul Stoddard Goldman Sachs & Co. LLC, Andrzej Tomczyk, CFA Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Pricing information

FedEx Corp. (Buy, \$315.62) and United Parcel Service Inc. (Buy, \$114.10)

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution				Investment Banking Relationships		
	Buy	Hold	Sell		Buy	Hold	Sell
Global	50%	34%	16%		65%	59%	43%

As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an

officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data

feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.